Gym Website (English/Urdu) - Project Documentation Pack

Version 1.0 | Status: Draft for approval | No coding starts until the PRD, TRD and architecture are approved.

This pack contains six documents: (1) PRD, (2) TRD, (3) App Flow, (4) UI/UX Design Brief, (5) Backend Schema, (6) Implementation Plan. Items marked **[A]** are assumptions and items marked **[Q]** are open questions for the gym owner. No real gym details, prices, reviews or certifications are invented; every such value is a placeholder to be supplied by the owner.

Recommended Project Scope

The full wish list in the master prompt (online payments, WhatsApp automation, attendance, full blog, reports) is too large for one release. The recommended approach is three releases.

- **MVP (Release 1):** Bilingual public website, plans and pricing display, trainers, class timetable, gallery, testimonials, FAQs, blog (simple), contact and trial-session inquiry forms, WhatsApp click-to-chat button, and a lean admin panel to manage content, plans, trainers, classes, and inquiries. Language switcher with full RTL/LTR support.
- **Release 2:** Member registration/login, member dashboard and profile, membership status and expiry tracking, manual payment recording by staff, class booking with capacity limits, email notifications, role-based staff permissions.
- **Release 3 (optional):** Online payments, attendance check-in, reports, audit log viewer, WhatsApp API automation, push reminders.

Why this scope: the site must first attract members (marketing value); operations features depend on real gym workflows that the owner must confirm [Q]. Building bookings and payments before those are confirmed risks rework.

Document 1: PRD (Product Requirements Document)

1. Executive Summary and Vision

A premium, responsive, fully bilingual (English LTR / Urdu RTL) gym website that converts visitors into trial-session leads and members, and gives gym staff a simple admin dashboard to run content and (from Release 2) memberships and bookings.

Vision: the most trusted, easy-to-use fitness website in the gym's local market, equally comfortable for English and Urdu readers.

2. Business Goals and Success Metrics

Baselines are unknown [Q]; targets below are proposals to be confirmed after launch measurement begins.

Goal	Metric	Proposed target
Generate leads	Trial/contact form submissions per month	Set baseline in month 1, then +25% by month 3
Convert visitors	Homepage-to-inquiry conversion rate	At least 3%
Serve Urdu users	Share of sessions in Urdu	Tracked; no target until baseline
Performance	Mobile Lighthouse Performance score	85 or higher on Home, Plans, Contact
Accessibility	Lighthouse Accessibility score	95 or higher
Reduce admin effort	Time to publish a new plan or class in both languages	Under 5 minutes
Reliability	Uptime	99.5% monthly

3. Target Users and Personas

- **Prospect (Ayesha/Ali, 18-40):** browses on mobile, wants prices, timings, trainers; may prefer Urdu. Goal: find a plan and book a trial quickly.
- **Member:** wants to see plan expiry, payment history, and book classes (Release 2).
- **Front-desk staff:** records members, payments, handles inquiries.
- **Trainer:** views own classes and bookings (Release 2).
- **Gym owner/admin:** controls content, pricing, staff roles, and sees metrics.

Persona names are illustrative only.

4. Scope: MoSCoW

Must have (MVP): responsive layout; EN/UR switcher with content and layout change; locale URLs; Home, About, Services, Plans, Trainers + detail, Classes + timetable, Gallery, Testimonials, Contact, FAQs, Blog list/detail, Privacy, Terms, 404; contact and trial inquiry forms with validation and spam protection; WhatsApp button; admin login; admin CRUD for plans, trainers, services, classes/timetable, gallery, testimonials, FAQs, blog, inquiries, site settings; bilingual content fields; SEO basics; accessibility basics; email notification to staff on new inquiry.

Should have (Release 2): member registration/login/password reset; member dashboard, profile, membership and payment history; booking and booking history; manual payment recording; membership expiry reminders by email; staff roles and permissions matrix; CSV export of inquiries and members.

Could have (Release 3): online payment gateway; attendance check-in; reports; audit log viewer; WhatsApp API messages; waitlists.

Future / Won't have now: native mobile app, store/e-commerce, diet plan engine, wearable integration, multi-branch support (unless owner confirms multiple branches [Q]).

5. Sitemap and Information Architecture

All public URLs are prefixed by locale: `/en/...` and `/ur/...`.

- Home
- About Us
- Services

- Plans and Pricing
- Trainers, then Trainer Detail
- Classes and Timetable
- Gallery
- Success Stories (Testimonials)
- Blog, then Article
- FAQs
- Contact (includes Trial Session form)
- Auth (R2): Login, Register, Forgot/Reset Password
- Member area (R2): Dashboard, Profile, Membership and Payments, Bookings
- Admin: Login, Dashboard, and management sections
- Legal: Privacy Policy, Terms, 404

6. Page-by-Page Specifications (summary)

Page	Key content	Key behavior / acceptance hint
Home	Header, hero with two CTAs (Join / Book Trial), intro, facilities, services, plans preview, featured trainers, timetable preview, testimonials, gallery, FAQ, contact, map, WhatsApp, footer	Every section driven by CMS content in both languages; LCP under 2.5s on mid-range mobile
About	Story, mission, facilities, team	Content owner-supplied; no invented claims
Services	Service cards with detail	Filter by category
Plans	Plan cards: name, duration, price, features, CTA	Currency formatted per locale; "Popular" badge optional
Trainers	Cards with photo, specialties, availability	Detail page with bio, schedule; certifications shown only if owner supplies proof [Q]
Classes	Weekly timetable and list	Filter by day, class type, trainer; RTL-aware table

Page	Key content	Key behavior / acceptance hint
Gallery	Image grid with lightbox	Lazy loaded; alt text in both languages
Testimonials	Real member stories only	Each entry needs member consent flag
Contact	Form, phone, WhatsApp, email, map, hours	Validation, honeypot, rate limit, success message in selected language
FAQs	Accordion	Searchable
Blog	List with search, categories, pagination; article page	Each post can exist in one or both languages
Auth (R2)	Forms with inline validation	Generic error messages to avoid account enumeration
Member area (R2)	Status card, expiry date, payments table, bookings	Only the member's own data
Admin	Metrics and CRUD screens	Server-side permission on every action

7. Functional Requirements (key list)

- **FR-01** Public pages render in the selected locale with translated content.
- **FR-02** Language switcher is visible in the header on all pages and swaps to the equivalent page in the other locale.
- **FR-03** Contact and trial forms store the inquiry, email staff, and show a localized confirmation.
- **FR-04** Admin can create/edit/publish/unpublish plans, trainers, services, classes, FAQs, gallery, testimonials, and blog posts, with separate English and Urdu fields.
- **FR-05** Admin inquiry list supports search, status filter (new, contacted, converted, closed), sorting, pagination.
- **FR-06** (R2) Members can register, verify email, log in, reset password.
- **FR-07** (R2) Membership has start date, end date, status (active, expiring, expired, frozen); status is computed from dates.

- **FR-08** (R2) Booking respects class capacity and cancellation cutoff (cutoff value [Q]).
- **FR-09** (R2) Staff can record payments (amount, method, reference, date); no card data stored.
- **FR-10** Admin roles restrict access per the permissions matrix.

8. Non-Functional Requirements

- Performance: LCP under 2.5s, CLS under 0.1, INP under 200ms on 4G mobile for key pages.
- Availability: 99.5% monthly.
- Browser support: last 2 versions of Chrome, Safari, Edge, Firefox; Android Chrome; iOS Safari.
- Accessibility: WCAG 2.2 AA.
- Scalability: comfortably supports 10,000 monthly visitors and 2,000 members on a single small deployment [A].
- Maintainability: no hardcoded user-facing strings in components; TypeScript strict mode; CI runs lint, type check, tests.
- Data: automated daily database backups, 30-day retention; restore tested before launch.

9. Localization, RTL and LTR Requirements

Locales: `en` (default, LTR) and `ur` (RTL) [A: default is English; owner to confirm].

Locale routing: every public route lives under `/[locale]/...`. Requests to `/` redirect to the saved locale, else to the browser's `Accept-Language` match, else to `en`. Admin and member areas use the same locale prefix.

Persistence: the chosen locale is saved in a `NEXT_LOCALE` cookie (1 year, SameSite=Lax) and, for logged-in users, in their profile. The URL always wins over the cookie, so shared links open in the intended language.

Switcher behavior: switching goes to the same page in the other locale (for example `/en/plans` to `/ur/plans`), preserving query parameters and scroll to top. If a blog post has no Urdu version, the switcher goes to the Urdu blog list and shows a notice.

Direction: the root `<html>` element gets `lang` and `dir` set from the locale (`lang="ur" dir="rtl"`). Styling uses CSS logical properties (margin-inline-start, padding-inline-end, text-align: start) and Tailwind logical utilities (ms-, me-, ps-, pe-, start-, end-) instead of left/right. Directional icons (arrows, chevrons, carousels) are mirrored in RTL; non-directional icons (play, search, clock) are not.

Typography: Urdu uses Noto Nastaliq Urdu for body and headings, or Noto Naskh Arabic if Nastaliq proves too tall for dense UI like tables and buttons [Q: owner preference]. Urdu line-height is increased (about 1.8-2.0 for Nastaliq). Fonts are self-hosted, subsetted, `font-display: swap`, with system fallbacks. English uses a bold athletic sans-serif (for example Inter plus a display face).

Numbers, prices, phones: numerals default to Western digits (0-9) in both locales for clarity [Q: owner may want Eastern Arabic-Urdu digits]. Prices are formatted with `Intl.NumberFormat` using the locale and currency (PKR assumed [A]). Phone numbers, emails, and URLs are wrapped in `<bdi>` or `dir="ltr"` spans so they do not reorder inside Urdu text.

Translation data structure:

- Static UI strings (buttons, labels, validation, errors, empty states): JSON message files per locale, namespaced (`common`, `nav`, `forms`, `errors`, `member`, `admin`), managed with a library such as next-intl. Keys are identical across locales; a CI check fails if a key is missing.
- Dynamic content (plans, trainers, blog, FAQs, etc.): stored in the database as paired fields or a translations table with `locale` column. Admin forms show English and Urdu side by side with a completeness indicator.
- Emails: separate templates per locale.

Adding new pages and content in both languages:

- New static page: add route under `[locale]`, add its message namespace in `en.json` and `ur.json`; CI blocks merge if either is missing.
- New CMS content: admin must fill both languages before "Publish" is enabled, unless the admin explicitly marks it "single language" (then it only appears in that locale).

Requirement IDs: L-01 locale prefixes; L-02 cookie persistence; L-03 html lang/dir set server-side; L-04 no physical left/right CSS; L-05 mirrored directional icons; L-06 fonts subsetted and total font payload under 300 KB per locale [A]; L-07 bidi-safe numbers/phones; L-08 hreflang tags on every public page; L-09 translation completeness CI check.

10. User Stories (with IDs)

ID	Release	Story	Priority
US-01	MVP	As a visitor, I want to switch between English and Urdu so I can read the site in my language.	Must
US-02	MVP	As a visitor, I want to see plans and prices so I can choose a membership.	Must
US-03	MVP	As a visitor, I want to request a free trial so the gym can contact me.	Must
US-04	MVP	As a visitor, I want to view trainers and their specialties.	Must
US-05	MVP	As a visitor, I want to see the class timetable and filter by day or type.	Must
US-06	MVP	As a visitor, I want to WhatsApp the gym in one tap.	Must
US-07	MVP	As a visitor, I want to find location, hours and phone quickly.	Must
US-08	MVP	As an admin, I want to edit plans, trainers, classes and content in both languages without a developer.	Must
US-09	MVP	As staff, I want to see and update the status of inquiries.	Must
US-10	MVP	As an admin, I want an email when a new inquiry arrives.	Must
US-11	R2	As a visitor, I want to register and verify my email.	Should
US-12	R2	As a member, I want to log in and reset a forgotten password.	Should
US-13	R2	As a member, I want to see my plan status and expiry date.	Should
US-14	R2	As a member, I want to book and cancel classes.	Should
US-15	R2	As a member, I want to see my payment and booking history.	Should

ID	Release	Story	Priority
US-16	R2	As front-desk staff, I want to record a payment and extend a membership.	Should
US-17	R2	As an owner, I want to assign roles so staff only see what they need.	Should
US-18	R3	As a member, I want to pay online.	Could
US-19	R3	As an owner, I want attendance and revenue reports.	Could
US-20	R3	As an owner, I want to review an audit log of admin actions.	Could

11. User Flows and Business Rules

Flows: see Document 3 (App Flow).

Business rules (assumptions to confirm [Q]):

- **BR-01** A membership's end date = start date + plan duration; status is derived (active if today is within dates, expiring if 7 days or fewer remain, expired otherwise).
- **BR-02** A member can hold only one active membership at a time [A].
- **BR-03** A class booking is allowed only for an active member and only while seats remain.
- **BR-04** Cancellation is free up to N hours before class (N to be set by the owner in settings).
- **BR-05** Payments are recorded as paid, pending, refunded, or failed; refunds are recorded, never deleted.
- **BR-06** Testimonials are published only with recorded member consent.
- **BR-07** Prices shown on the site come only from the plans table; no hardcoded prices.
- **BR-08** Deleting records with financial links is blocked; they are archived (soft delete).

12. UI/UX Design System

Summarized here; full detail is in Document 4.

13. Technical Architecture

Summarized here; full detail is in Document 2 (TRD).

14. Database Schema

Summarized here; full detail is in Document 5 (Backend Schema).

15. API Requirements (summary)

- REST-style route handlers under `/api`, JSON only, validated by schemas (Zod).
- Public: `POST /api/inquiries`; `GET` endpoints for published content are served through server components, not exposed unless needed.
- Auth (R2): register, verify email, login, logout, forgot/reset password.
- Member (R2): `GET/PATCH /api/me`, `GET /api/me/membership`, `GET /api/me/payments`, `GET/POST/DELETE /api/bookings`.
- Admin: CRUD under `/api/admin/{resource}` with pagination, sorting and filters via query params.
- Standard error shape: `{ error: { code, message_key, details } }`, where `message_key` is translated client-side.
- Rate limits: inquiries 5/hour/IP; login 10/15 min/IP+account.

16. Admin Roles and Permissions Matrix

Resource	Owner	Manager	Front Desk	Trainer	Content Editor
Dashboard metrics	Full	Full	Limited	Own	None
Members	Full	Full	Create/Edit	Own assigned (view)	None
Memberships and plans	Full	Full	Assign/renew	View	None
Payments	Full	Full	Record	None	None
Classes, schedules	Full	Full	View	Own (edit notes)	None
Bookings, attendance	Full	Full	Full	Own classes	None
Trainers, services	Full	Full	View	Own profile	Edit
Inquiries	Full	Full	Full	None	None

Resource	Owner	Manager	Front Desk	Trainer	Content Editor
Gallery, testimonials, blog, FAQs	Full	Full	None	None	Full
Site settings, translations	Full	Edit	None	None	Edit translations
Staff roles	Full	None	None	None	None
Audit logs	Full	View	None	None	None

Permissions are enforced on the server for every request; hiding a button is never the only control.

17. Security and Performance Requirements

- Passwords hashed with Argon2id or bcrypt; never logged.
- Sessions in HttpOnly, Secure, SameSite=Lax cookies; CSRF protection on state-changing requests.
- Server-side validation on all inputs; output encoding to prevent XSS; parameterized queries only (ORM) to prevent SQL injection.
- Rate limiting and honeypot/CAPTCHA on public forms and login.
- File uploads: allow-list of image types, max 5 MB, server-side type check, re-encoded, stored in object storage with random names.
- Security headers (CSP, HSTS, X-Content-Type-Options, Referrer-Policy).
- Payment data: raw card details are never stored or processed by our servers; a hosted gateway checkout is used in R3.
- Privacy: consent checkbox on forms, data export/delete on request, retention rules in Document 5.
- Backups: daily automated, 30-day retention, restore drill before launch.
- Performance: image optimization (WebP/AVIF), responsive sizes, lazy loading, static generation with revalidation for public pages, CDN caching.

18. SEO and Accessibility Requirements

- Unique title and meta description per page per locale; `hreflang` pairs and `x-default`; canonical URLs; sitemap.xml including both locales; robots.txt.

- Open Graph and Twitter cards; JSON-LD (`LocalBusiness` / `HealthClub`, `Article`, `FAQPage`) using only owner-supplied facts.
- WCAG 2.2 AA: contrast of lime on charcoal verified (at least 4.5:1 for text), visible focus states, keyboard navigation, skip link, form labels and error association, alt text in both languages, reduced-motion support, screen-reader announcement of language change.

19. Testing Plan and Acceptance Criteria

Testing layers: unit (Vitest), component (Testing Library), end-to-end (Playwright, run in both locales and on mobile viewport), accessibility (axe), performance (Lighthouse CI), security (dependency audit, basic OWASP checks), manual RTL review by a native Urdu reader.

Sample acceptance criteria (Given/When/Then):

- **AC-01 (US-01)** Given I am on `/en/plans`, when I click the Urdu switcher, then I land on `/ur/plans`, the page is rendered with `dir="rtl"`, all UI text and plan content appear in Urdu, and a refresh keeps Urdu.
- **AC-02 (US-01)** Given I previously chose Urdu, when I open `/` in a new session, then I am redirected to `/ur`.
- **AC-03 (US-03)** Given I fill valid name, phone and consent, when I submit the trial form, then an inquiry is saved, staff receive an email, and I see a success message in my language within 3 seconds.
- **AC-04 (US-03)** Given I submit with an invalid phone number, when validation runs, then I see an inline localized error and no record is created.
- **AC-05 (US-02)** Given a plan is unpublished in admin, when a visitor loads the plans page, then that plan is not shown.
- **AC-06 (US-08)** Given an admin leaves the Urdu title empty, when they click Publish, then publishing is blocked with a message unless "single language" is selected.
- **AC-07 (security)** Given a Front Desk user, when they call a staff-role API directly, then the server returns 403.
- **AC-08 (R2, US-14)** Given a class is full, when a member tries to book, then the booking is refused with a localized message.
- **AC-09 (RTL)** Given the Urdu locale, when any page is viewed at 360px width, then there is no horizontal scroll and phone numbers display left-to-right intact.

20. Risks, Dependencies and Open Questions

Risk	Impact	Mitigation
Urdu translations are inaccurate or machine-only	Credibility	Native-speaker review before launch
Nastaliq font is heavy and tall	Speed, layout	Subset font; test Naskh alternative
Owner does not supply real photos, prices, trainer data	Launch delay	Content checklist with deadlines; placeholders blocked from production
Scope creep into payments/attendance	Cost, time	Phase gating; change control
Payment gateway availability/approval	R3 delay	Start merchant onboarding early; manual payments in R2
Spam inquiries	Staff noise	Honeypot, rate limit, CAPTCHA

Open questions [Q]: gym name, logo, brand colors beyond lime/charcoal; number of branches; real plan names and prices; currency; operating hours; trainer list and credentials; class list and capacities; cancellation policy; are member accounts and bookings needed at launch; payment methods in use; default language; Urdu digit style; domain and hosting preference; who writes and translates content; legal pages source.

Dependencies: domain, hosting account, email sender domain, map API or embed, WhatsApp business number, real photography, translated content.

Roadmap and Definition of Done

See Document 6. **Definition of Done (any feature):** requirements and acceptance criteria met in both locales; responsive at 360, 768, 1280px; WCAG AA checks pass; tests written and passing in CI; no hardcoded strings; code reviewed; deployed to staging and approved; documentation updated.

Document 2: TRD (Technical Requirements Document)

1. Recommended Architecture

A single Next.js (App Router) full-stack application with TypeScript, deployed as one unit. This avoids a separate backend service: for a gym site of this size, a monolith is cheaper, simpler to maintain, and fast enough.

Layer	Choice	Reason
Framework	Next.js 15 (App Router), React 19	SSR/SSG for SEO, route handlers for APIs, mature i18n support
Language	TypeScript (strict)	Fewer runtime bugs, shared types
Styling	Tailwind CSS with logical utilities	Fast, consistent, easy RTL via `ms-/me-/ps-/pe-`
i18n	next-intl	Locale routing, ICU messages, server components support
Database	PostgreSQL	Relational integrity for members, plans, payments, bookings
ORM	Prisma or Drizzle	Typed queries, migrations
Auth	Auth.js (NextAuth) with credentials + email verification, or Better Auth; Argon2id hashing	Avoids building auth from scratch
Validation	Zod	One schema for client and server
Forms	React Hook Form + Zod resolver	Accessible, performant
Images	Cloudinary, or S3-compatible storage (Cloudflare R2) plus Next/Image	Optimization and CDN
Email	Resend, Postmark, or SMTP provider	Transactional emails per locale

Layer	Choice	Reason
Hosting	Vercel (app) + managed Postgres (Neon, Supabase, or similar)	Low ops; alternatives: VPS with Docker if owner prefers
Rate limiting	Upstash Redis or database-backed limiter	Protect forms and login
Monitoring	Sentry + uptime monitor	Error visibility
Testing	Vitest, Testing Library, Playwright, axe	Covered in PRD

Specific versions and vendors are recommendations to be verified at project start; availability and pricing of vendors in Pakistan [Q] should be checked before final selection.

Why not alternatives: a separate REST backend (NestJS/Express) adds deployment and sync cost without MVP benefit; a headless CMS (Strapi, Sanity) is viable but adds a second system; the built-in admin panel with bilingual fields is simpler. Revisit if non-technical editors need richer editing workflows.

2. System Context

Browser (EN/UR) to CDN to Next.js app (server components, route handlers) to PostgreSQL, object storage, email service, and (R3) payment gateway and WhatsApp API.

3. Project Structure

- `app/[locale]/(public)/...` public pages
- `app/[locale]/(member)/...` member area (R2)
- `app/[locale]/admin/...` admin panel
- `app/api/...` route handlers
- `messages/en.json`, `messages/ur.json` UI strings
- `lib/` db client, auth, permissions, i18n helpers, email
- `components/` reusable UI (Button, Card, Table, FormField, LanguageSwitcher)
- `prisma/` schema and migrations
- `tests/` unit and e2e

4. Localization Implementation

1. Middleware detects locale: URL prefix, then `NEXT_LOCALE` cookie, then `Accept-Language`, then default `en`.
2. Root layout sets `<html lang dir>` from locale on the server (no flash of wrong direction).
3. Messages are loaded per request by namespace; client bundles receive only needed namespaces.
4. Dynamic content is read from the database with a locale filter and an English fallback only if explicitly allowed per field (otherwise the item is hidden in that locale).
5. Fonts loaded with `next/font/local`, subsetted per locale, applied via a CSS variable on `html[lang=ur]`.
6. Tailwind: use logical properties; add `rtl:` variants only where logical utilities are insufficient (for example icon mirroring `rtl:-scale-x-100`).
7. Dates and numbers via `Intl` with locale; calendar is Gregorian [A].
8. SEO: `alternates.languages` metadata for hreflang.
9. CI script compares key sets of `en.json` and `ur.json` and fails on mismatch or empty values.

5. Authorization Design

- Roles: owner, manager, front_desk, trainer, content_editor, member.
- A central `can(user, action, resource)` helper reads a permissions map (database-backed in R2) and is called in every route handler and server action.
- Row-level checks: members only access rows where `user_id` matches; trainers only their classes.
- Admin routes also guarded in middleware, but middleware is never the only check.

6. Non-Functional Technical Requirements

- **TR-01** TypeScript strict; ESLint and Prettier enforced in CI.
- **TR-02** All env secrets in environment variables; `.env.example` committed, real secrets never.
- **TR-03** Public pages statically generated with on-demand revalidation when admin publishes changes.

- **TR-04** Images served as WebP/AVIF with explicit width/height to prevent layout shift.
- **TR-05** API responses under 300 ms p95 for typical reads [A].
- **TR-06** Database migrations are versioned and reversible; no manual schema edits in production.
- **TR-07** Structured logging with no personal data or secrets.
- **TR-08** Three environments: local, staging, production.
- **TR-09** CI pipeline: install, lint, type check, unit tests, build, e2e on staging.
- **TR-10** Backups and restore procedure documented and rehearsed.

7. Third-Party Integrations

Integration	Release	Notes
Google Maps embed or OpenStreetMap	MVP	Embed only; no API key if possible
WhatsApp click-to-chat (`wa.me` link)	MVP	No API needed
Email provider	MVP	SPF/DKIM configured
Analytics (privacy-friendly)	MVP	Consent banner if required
Payment gateway	R3	Choose after owner confirms accepted methods; hosted checkout only
WhatsApp Business API	R3	Requires approval and templates

Document 3: App Flow (User and Admin Experience)

1. Visitor Flows

Flow A: Language selection

1. Visitor lands on `/`.
2. System checks cookie, then browser language, then defaults to English.
3. Redirect to `/en` or `/ur`; page renders with correct direction.

4. Visitor clicks switcher; same page loads in other locale; cookie updated.

Flow B: Discover and request a trial

1. Home hero CTA "Book a Free Trial" (label text to be confirmed by the owner) opens the Contact/Trial form.
2. Visitor enters name, phone, optional email, preferred time, message, consent.
3. Client validation, then server validation, honeypot and rate limit.
4. Success: inquiry saved, staff emailed, confirmation shown (and optionally a visitor email).
5. Failure: inline localized error; data preserved in the form.

Flow C: Choose a plan Home or Plans page, compare plan cards, click "Join" which goes to Contact with plan preselected (MVP) or to Register/Checkout (R2/R3).

Flow D: Explore Trainers list to Trainer detail to Classes filtered by that trainer to Trial form.

Flow E: Contact by WhatsApp Floating button opens `wa.me` with a prefilled localized message.

2. Member Flows (Release 2)

Registration: Register, submit form, verification email, click link, account active, redirected to dashboard.

Login / reset: Login with email and password; failed attempts are rate-limited; Forgot Password sends a time-limited single-use link; reset sets a new password and invalidates other sessions.

Dashboard: shows membership status chip, days remaining, next booking, recent payments, quick actions.

Book a class: Classes timetable, select class, system checks active membership and capacity, confirm, email confirmation. Cancel before cutoff releases the seat.

Profile: edit name, phone, emergency contact, preferred language, password.

3. Admin Flows

Login: admin login page, credentials, optional 2FA (R3), role loaded, redirected to dashboard.

Publish bilingual content: open resource (for example Plan), fill English tab and Urdu tab, upload image with alt text in both languages, preview in each locale, Publish; the system triggers revalidation of affected public pages.

Handle inquiry: inquiry list shows New; staff opens, calls or WhatsApps, adds a note, sets status Contacted/Converted/Closed.

Register a member and payment (R2): Create member, assign plan, record payment, system calculates end date, member receives email.

Renewals (R2): list of members expiring within 7 days; staff records a renewal payment which extends the membership.

4. Key Screen States

Every screen defines: loading, empty, error, success, and no-permission states, each with translated copy.

Document 4: UI/UX Design Brief

1. Design Direction

Premium, athletic, high-energy but clean. Dark charcoal base, lime-green accent, bold typography, large photography, generous spacing. Brand name, logo, and photography come from the owner [Q]; no stock claims about the gym are made.

2. Design Tokens (initial proposal)

Token	Value (proposal)	Use
Background	#0F1115	Page
Surface	#181B21	Cards
Surface raised	#222630	Hover, modals
Primary accent	#B6F500 (lime)	CTAs, highlights
Text primary	#F5F7FA	Body
Text muted	#A3A9B5	Secondary

Token	Value (proposal)	Use
Border	#2C313B	Dividers
Danger / Success / Warning	#FF5C5C / #3DDC84 / #FFB020	Status

Lime on charcoal has strong contrast; text on a lime button must be dark (#0F1115). All contrast pairs are verified during design (at least 4.5:1 for body text). A light-theme variant is a Could-have.

Spacing: 4px base scale (4, 8, 12, 16, 24, 32, 48, 64, 96). **Radius:** 12px cards, 999px pills. **Elevation:** subtle borders over heavy shadows.

3. Typography

- English: bold geometric or condensed display font for headings (for example Oswald or Barlow Condensed) and Inter for body.
- Urdu: Noto Nastaliq Urdu (headings and body) with Noto Naskh Arabic as fallback or UI-dense alternative; larger line-height; avoid letter-spacing and uppercase transforms in Urdu.
- Type scale: 14, 16, 18, 20, 24, 32, 48, 64 (fluid on desktop). Urdu sizes are about 10 percent larger for equivalent legibility.

4. Components

Header with logo, nav, language switcher (visible at all breakpoints, shows the target language's native name: "English" / "اردو"), CTA button; mobile drawer menu; buttons (primary, secondary, ghost); plan card; trainer card; class card and timetable; testimonial card; gallery grid and lightbox; accordion; form fields with error text; toast; table with sort/filter/pagination; status chips; modal; empty state; skeleton loaders; floating WhatsApp button (placed at inline-end corner so it mirrors in RTL).

5. RTL Mirroring Rules

- Whole layout mirrors: nav order, grid flow, sidebars, breadcrumbs, progress bars, timelines.
- Mirror: arrows, chevrons, back/forward, carousels, list bullets, slide direction.

- Do not mirror: logos, photos, play icons, clocks, media controls, charts' numeric axes, phone numbers, code.
- Mixed content: wrap English brand names, numbers, emails and phone numbers in bidi-isolated spans.
- Test all components in both directions in the component library (Storybook optional) before page assembly.

6. Responsive Breakpoints

Mobile 360-639, tablet 640-1023, desktop 1024-1439, wide 1440 and above. Mobile-first. Tap targets at least 44x44px.

7. Accessibility and Motion

Visible focus ring (lime outline), keyboard-operable switcher and menu, semantic landmarks, `prefers-reduced-motion` honored, no information conveyed by color alone, alt text in both languages.

8. Imagery

High-quality real photos of the actual gym, equipment and trainers (with consent). Dark overlays on hero for text legibility. Do not use stock images presented as the gym's own facilities.

9. Admin UI

Sidebar (collapses on mobile, mirrors in RTL), top bar with language toggle and profile, dense but readable tables, bilingual form layout using English/Urdu tabs with completeness badges, sticky Save/Publish bar, confirm dialogs for destructive actions.

10. Deliverables

Low-fidelity wireframes for Home, Plans, Trainers, Classes, Contact, Admin Dashboard in both directions; high-fidelity mockups after wireframe approval; clickable prototype for the main flows; design tokens exported to Tailwind config.

Document 5: Backend Schema

PostgreSQL. Conventions: UUID primary keys (`id`), `created_at` and `updated_at` timestamps (timestamptz) on every table, soft delete via `deleted_at` where noted. Entities marked R2/R3 are created only when that release starts. Translatable content uses a companion `*_translations` table keyed by `(parent_id, locale)`.

1. Core Tables (MVP)

users (admin staff in MVP; members added in R2)

- id, email (unique, lowercase), password_hash, role (enum: owner, manager, front_desk, trainer, content_editor, member), preferred_locale (en/ur), email_verified_at, is_active, last_login_at, created_at, updated_at.
- Indexes: unique(email), index(role).

services and **service_translations**

- services: id, slug (unique), icon, image_id, sort_order, is_published.
- service_translations: service_id FK, locale, title, description; unique(service_id, locale).

membership_plans and **membership_plan_translations**

- plans: id, slug (unique), price_amount (integer, minor units), currency (default per settings), duration_days, is_popular, is_published, sort_order, deleted_at.
- translations: plan_id FK, locale, name, description, features (jsonb array of strings); unique(plan_id, locale).
- Constraint: price_amount >= 0, duration_days > 0.

trainers and **trainer_translations**

- trainers: id, slug (unique), photo_id, phone (optional), email (optional), is_published, sort_order, user_id (nullable FK, R2).
- translations: trainer_id, locale, full_name_display, bio, specialties (jsonb); certifications only as owner-verified text.
- **trainer_availability**: id, trainer_id FK, weekday (0-6), start_time, end_time. Check: end_time > start_time.

class_types and **class_type_translations**

- class_types: id, slug, image_id, default_capacity, is_published.

- translations: class_type_id, locale, name, description.

class_sessions (recurring timetable entries)

- id, class_type_id FK, trainer_id FK, weekday, start_time, end_time, capacity, room (text), is_active.
- Indexes: (weekday, start_time), trainer_id. Check: end_time > start_time, capacity > 0.

inquiries

- id, type (enum: contact, trial), full_name, phone, email (nullable), preferred_time (nullable), plan_id (nullable FK), message, locale, consent_given_at, status (new, contacted, converted, closed), assigned_to (nullable FK users), internal_note, ip_hash, created_at, updated_at.
- Indexes: (status, created_at desc), phone.
- Retention: delete or anonymize closed inquiries after 24 months [A].

testimonials and **testimonial_translations**

- testimonials: id, member_display_name, photo_id (nullable), rating (1-5, nullable), consent_recorded_at (required to publish), is_published.
- translations: testimonial_id, locale, quote.

gallery_images

- id, media_id FK, category, sort_order, is_published; **gallery_image_translations**: image_id, locale, alt_text, caption.

media

- id, storage_key, url, mime_type, width, height, size_bytes, uploaded_by FK users.

faqs and **faq_translations**

- faqs: id, category, sort_order, is_published; translations: faq_id, locale, question, answer.

blog_posts and **blog_post_translations**

- posts: id, slug (unique), cover_media_id, author_id FK users, status (draft, published), published_at.
- translations: post_id, locale, title, excerpt, body (sanitized rich text or markdown), seo_title, seo_description; unique(post_id, locale).
- Index: (status, published_at desc). Full-text index on translations for search.

site_settings

- key (primary key), value (jsonb), updated_by. Holds gym name per locale, address per locale, phone, WhatsApp number, email, opening hours, map coordinates, social links, default currency, cancellation cutoff hours. Contact facts are owner-supplied.

ui_translations (optional)

- id, namespace, key, locale, value; unique(namespace, key, locale). Only if non-developers must edit UI strings; otherwise JSON files in the repo are the source of truth [Q].

2. Release 2 Tables

member_profiles

- user_id (PK, FK users), full_name, phone, date_of_birth (nullable), gender (nullable), emergency_contact_name, emergency_contact_phone, member_code (unique), joined_at, notes (staff only).
- Personal data minimized: collect only what the owner confirms is needed [Q].

memberships

- id, member_id FK users, plan_id FK, start_date, end_date, status (active, expired, frozen, cancelled), created_by FK, created_at.
- Constraint: end_date >= start_date. Partial unique index to enforce one active membership per member (BR-02).
- Index: (member_id, status), end_date (for expiry reminders).

payments

- id, member_id FK, membership_id FK (nullable), amount (integer minor units), currency, method (enum: cash, bank_transfer, card, wallet, other), status (paid, pending, failed, refunded), reference (text, nullable), paid_at, recorded_by FK, notes.
- Never stores card numbers, CVV or full account numbers; only gateway reference IDs in R3.
- Indexes: member_id, paid_at, status. Check: amount > 0 except refund rows.

bookings

- id, member_id FK, class_session_id FK, session_date (date), status (booked, cancelled, attended, no_show), booked_at, cancelled_at.
- Unique (member_id, class_session_id, session_date) to block duplicates. Capacity enforced in a transaction counting booked rows per (class_session_id, session_date).

- Indexes: (class_session_id, session_date), member_id.

notifications

- id, user_id FK, channel (email, whatsapp), template_key, locale, status (queued, sent, failed), payload (jsonb, no secrets), sent_at, error.

password_reset_tokens / email_verification_tokens

- id, user_id FK, token_hash (never the raw token), expires_at, used_at. Single-use; expire in 1 hour (reset) and 24 hours (verify).

sessions (if the auth library requires): handled by the library.

3. Release 3 Tables

- **attendance**: id, member_id, class_session_id, checked_in_at, method.
- **audit_logs**: id, actor_user_id, action, entity_type, entity_id, before (jsonb), after (jsonb), ip_hash, created_at; append-only; index (entity_type, entity_id), (actor_user_id, created_at). Retention 12-24 months [A].
- **payment_gateway_events**: id, provider, event_id (unique for idempotency), payload, processed_at.
- **permissions** (if configurable roles are required): role, resource, action.

4. Key Relationships

- users 1-1 member_profiles; users 1-many memberships, payments, bookings.
- membership_plans 1-many memberships.
- memberships 1-many payments.
- trainers 1-many class_sessions; class_types 1-many class_sessions.
- class_sessions 1-many bookings.
- Every parent with translatable text has 1-many `*_translations` rows, one per locale.
- media referenced by trainers, services, gallery, blog, testimonials.

5. Data Integrity and Retention Rules

- Foreign keys use ON DELETE RESTRICT for financial records; content translations use ON DELETE CASCADE.
- Money stored as integers in minor units; no floating point.

- Publish rule: a record is visible in a locale only if its translation row exists and is complete (enforced in application and verified by tests).
- Personal data: inquiries 24 months after closure; member data retained while membership exists plus 24 months after expiry [A]; payments kept for the period required by applicable law, to be confirmed with the owner's accountant [Q]; backups encrypted, 30-day rotation; data export and deletion on member request, with financial records anonymized rather than removed.

Document 6: Implementation Plan

Durations assume one full-stack developer plus part-time designer; adjust for actual team size [A]. Content delivery by the owner is the most common delay.

Phase 0: Discovery and Approval (1 week)

- Confirm scope (MVP vs Release 2 at launch), answers to open questions, brand assets, domain, hosting.
- Sign off PRD, TRD, schema, and design brief.
- **Exit:** written approval; content checklist handed to the owner.

Phase 1: Design (1-2 weeks, overlaps Phase 2)

- Wireframes in LTR and RTL; tokens; component sheet; mockups for Home, Plans, Trainers, Classes, Contact, Admin; prototype review.
- **Exit:** approved designs and Urdu typography decision.

Phase 2: Foundation (1 week)

- Repository, CI, environments, Next.js + Tailwind + TypeScript setup, i18n and locale routing, RTL base layout, font loading, database and migrations, auth for staff, role helper, error and logging setup.
- **Exit:** blank bilingual shell deployed to staging; language switcher works with persistence; CI translation-key check runs.

Phase 3: Public Website (2-3 weeks)

- Reusable components, then pages: Home, About, Services, Plans, Trainers, Classes, Gallery, Testimonials, FAQs, Blog, Contact, legal, 404.
- Inquiry form with validation, spam protection, email notification; WhatsApp button; map; SEO metadata and sitemap; image optimization.
- **Exit:** all MVP public pages complete in both languages with placeholder or owner content.

Phase 4: Admin MVP (2 weeks)

- Admin layout; CRUD with bilingual forms for plans, trainers, services, classes, gallery, testimonials, FAQs, blog; inquiry management; settings; media upload; publish and revalidation.
- **Exit:** owner can update all content without a developer.

Phase 5: Quality and Launch (1-2 weeks)

- E2E tests in both locales; accessibility audit; Lighthouse tuning; security checklist; native Urdu proofreading; content entry; backup and restore test; DNS, SSL, analytics, monitoring; launch checklist; owner training (short guide in English and Urdu).
- **Exit:** Definition of Done met; go-live approval.

MVP total estimate: about 8-11 weeks [A].

Release 2 (6-8 weeks, after MVP approval)

Member auth and profile, memberships, manual payments, bookings with capacity rules, email notifications and expiry reminders, staff roles matrix, member and booking admin screens, CSV exports, expanded tests.

Release 3 (4-6 weeks, optional)

Payment gateway integration, attendance, reports, audit log viewer, WhatsApp API, 2FA for admins.

Maintenance after Launch

Monthly dependency updates, backup checks, uptime monitoring, content updates, analytics review, quarterly security review.

Roles and Responsibilities

Role	Responsibility
Gym owner	Business details, content, photos, approvals, Urdu content review
Product/Project manager	Scope control, timelines, acceptance
Designer	UX/UI, RTL design
Developer	Implementation, tests, deployment
Urdu proofreader	Translation quality

Immediate Next Steps

1. Owner answers the open questions listed in the PRD (Section 20).
2. Approve scope: MVP only, or MVP plus member accounts at launch.
3. Approve the technology stack and hosting choice.
4. Start design and content collection in parallel.